

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 5

Symmetry and Invariance: Why Some Operations Are Powerful

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. You discover that a dataset has the same statistical properties when you reverse the time axis. What does Noether's theorem (or its discrete analogue) suggest you should look for?

2. A colleague proposes solving a complex problem by brute-force simulation. You suspect there is a faster analytical solution. Where, specifically, would you look for the symmetry that would make it possible?

3. Why is the Fourier transform powerful for analyzing audio signals but not for analyzing images of faces?

4. A biologist says: “This protein’s function is conserved across species.” A physicist says: “Energy is conserved in a closed system.” Are these two uses of “conserved” related, or is this an accidental coincidence of language?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
